Manual

Optimization of Assignments (Occupancy)

HLS-BOP 8.2

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Contents

1 Optimization of Assignments (Occupancy)

Integration and functions

The "optimization" function has been integrated in the graphic planning board and provides the following, advanced functions:

- Optimization algorithm based on evolutionary strategies for which parameters are varied (different weighting) to perform several planning runs and the best parameters are used for final planning.
- Configuration options to define optimization parameters.
- Selection of pre-defined key figures/KPIs (basic key figures) to specify optimization.
- Definition of individual KPIs by combining and weighting the basic key figures existing in HYDRA.
- Definition of which key figure is to be optimized (e.g. optimization of the order processing time).
- Specification of which weighting parameters (e.g. processing time, priority) are to affect planning and how often weighting parameters are to be varied (iterations of planning).
- Automatic creation of individual plans by varying the defined weighting parameters and assessments based on the specified KPIs.
- Final planning based on the best weighting parameters and integration of the plan to the planning board. Presentation of the KPIs resulting from final planning.

2 Optimization

Overview

Menu	Production control → Preparations for production → Graphic planning
Tab	Optimization
Function authorization	sfs.opt

Purpose

The "optimization" function of the HYDRA Shop Floor Scheduling uses configurable targets to identify the best assignment/occupancy. To identify the best assignment/occupancy, several planning runs are performed. The user specifies the number of planning runs.

Each planning run performs an automatic assignment. For each run, a key figure is calculated that evaluates the planning. The separate planning runs use different input parameters to vary the planning ("mutation"). The optimization algorithm of the HYDRA Shop Floor Scheduling uses the planning strategy of the *target-oriented planning*. Using this strategy, the different planning parameters can be changed.

The parameters that influence the automatic assignment are changed at random. The different planning results are evaluated using key figures defined by the user. The key figures are calculated at the end of each planning run.

When all planning runs are performed, the planning run with the best key figure is identified. The parameters used in this planning run are then used for the final planning.

Requirements

Load a planning scenario in the Graphic planning to perform an optimization run.

You must be authorized to use the optimization function (license HLS-BOP) and the functions of the detailed planning and assignment (HLS-FBF).

Toolbar

Tab *Optimization* in the toolbar provides the different optimization functions. This tab is only visible if the relevant license is available.

Start optimization

Function authorization: sfs.opt

Opens the dialog to enter optimization parameters and to perform the optimization..

Show last result

Shows the last optimization result.

Key figures

Opens the configuration of Key figures.

Start optimization

To perform an optimization run, you must load a planning scenario in the planning board.

The optimization process of HYDRA Shop Floor Scheduling can be divided into the following steps:

1) Definition of parameters

Make the following settings in the *Optimization* dialog for an optimization run.

Optimization based on key figure

Select the key figure used for the optimization. The objective of different planning processes in an optimization run is to identify the best possible value for this key figure.

For optimization, you can use one of the key figures that can also be displayed in the graphic planning board (Gantt). You can use a basic key figure or a key figure defined by the user.

If the dialog is opened for the first time or in case the relevant file has been deleted in the local log directory, the last key figure included in the list is used by default.

Parameters

To vary the different planning runs ("mutation of characteristics"), the parameters of the planning strategy "target-oriented planning" are used. These parameters get a different weighting in each of the planning runs. The weighting is defined at random for each planning run. The user defines the range for the weighting of a parameter, i.e. the minimum and maximum variation.

The total of all weightings of the different parameters can be greater or less than 100. In this case, weightings are converted internally to the basis 100 [%].

Number of iterations per parameter

Planning processes are performed for each parameter having a weighting > 0. This value specifies how often the parameter is changed (number of variations). A maximum of 999 planning processes (iterations) can be performed for each parameter. The total number of planning runs is calculated as follows:

Number of planning processes = (number of parameters with weighting > 0) x (number of iterations per parameter) + 1.

The last planning (+ 1) is based on the parameters of the key figure with the best result (see below).

The more iterations are made per parameter, the longer the optimization time!

Subject to the number of iterations and number of set parameters having a weighting > 0, the duration of an optimization run can range between a few minutes and several hours! During that time, you cannot work on the MOC.

We recommend to perform optimization runs with only a few iterations (e.g. 10) and to monitor the run time.

Also watch the optimization result. Because it is possible that a high number of iterations does not lead to better results.

2) Performing planning runs

When you have entered the required parameters and the dialog is confirmed, the optimization is performed. For the optimization, the different planning runs are performed (simulated). A dialog shows the planning progress.

You cannot cancel an optimization run!

We recommend to perform optimization runs with only a few iterations (e.g. 10) and to monitor the run time.

To make sure that you can compare the different planning results, the planning always uses the same initial assignment/occupancy situation. This means: All operations that are not fixed are deallocated and a complete re-planning can be performed.

A "coefficient of sequence" (RFK) is specified for each operation at the beginning of each planning. This coefficient is calculated using the selected parameters (parameters with weighting > 0) and specifies the sequence ("queue") used to plan the different operations.

Example

RFK = K.BearbZ * BearbZ + K.PufferZ * PufferZ + K.PRIO * Prio

K.x are the weighting factors that specify the influence of the separate elements.
BearbZ, PufferZ, Prio are values of the order or operation. These values are standardized internally to make sure that the values can be compared (e.g. prio 0 - 9 does not match a processing time (BearbZ) of 1000 h).

The individual plans are performed for each parameter (with the specified number of iterations per parameter). In this context, the weighting factor varies randomly within the range resulting from the specified variation (see above). The assignments are made according to the same principles as for a simple, automatic assignment. The key figure (see above) defined by the planner is identified and saved at the end of each planning run.

The sequence used to vary the different weighting parameters is as follows:

- Processing time
- Buffer time
- Lateness
- Reduction level
- Priority
- Number of capacities

The different weighting parameters are explained below.

When all planning processes for the different parameters are completed, the weighting factor is identified that returned the best key figure. This weighting factor is used for further planning.

3) Running the best plan

When all planning processes with all parameters are performed, the different planning processes are evaluated and the "best planning" is identified. The "best planning" is the planning with the best result for the key figure specified by the planner.

The final planning is then performed using the parameters and weightings of the "best planning" identified before.

At the end of the optimization run, a dialog "optimization result" opens showing the results of the optimization run. The dialog is described in section "Show last result“.

When the last planning is executed, this planning is not immediately stored in the database. You can reject the planning without saving or edit the planning and then save it.

Show last result

You can only call this function if you have performed at least one optimization run.

It can take several minutes until the dialog is displayed that shows the results of the (last) optimization run. The time depends on the number of iterations and the number of set parameters having a weighting > 0.

The dialog "optimization result" shows the results of the current or last optimization run. The dialog shows the following information:

Progression of key figures

A line chart shows the progression of the key figure that has been specified at the beginning of the (last) optimization run.

For each iteration, the optimization function shows the key figure without dimension that has internally been identified.

The x-axis shows the numbers of plannings included in the optimization run. The y-axis is not labeled because the values are displayed without dimension.

Distribution of optimized parameters in percent

The parameter values of the planning run that has achieved the best key figure is displayed to the right of the line chart.

The absolute values are shown in the lower section; the upper section shows the distribution of these values in a pie chart. The values are standardized based on 100%. Because these standardized values deviate from the absolute values, they are not labeled in the pie chart.

The information displayed here is stored locally in files of the user directory c:\Users\<user>\AppData\Roaming\MPDV\MOC\log\ during the optimization run. If this directory or its files are deleted, a last result can no longer be displayed.

Weighting parameters

The following weighting parameters can be selected:

Processing time

The processing time is calculated using the remaining run time of the operation according to the remaining run time formula plus (static) setup and teardown time. You use the option "Prioritize longer processing time" to specify whether operations with shorter processing times or operations with longer processing times take priority.

If an operation can be planned for several, alternative workplaces, the average of all processing times of the workplaces is used as processing time.

Buffer time

Buffer time of the order (order buffer).

Delay time

The delay time is calculated from the scheduled end of the order minus the basic end date of the order.

Reduction level

Current reduction level of the order (only reasonable if reduction strategies are used).

Priority

Priority of order.

Number of alternative capacities

The number of alternative (primary) capacities results from the available resources that could be used. If production variants are available, these variants are also integrated.

Example for variability

Parameter	Weighting	Variability
Processing time	30	20
Delay time	70	50
Number of iterations:	5	

Using the above sample figures and a weighting of 30 [%] for the criterion "processing time", then the following range is calculated with 5 deviation values specified randomly.